Network File Systems – [NFS]

❖ What is NFS? [Port NFS - 2049]

➢ NFS stands for Network File System. It's a **distributed file system protocol** that allows users to access files over a network as if they were on their local **hard drives.**

✓ **Key Points:**

- **Developed by**: Sun Microsystems in 1984.
- **Used for**: Sharing files and directories between systems over a network.
- **Common in**: Unix/Linux environments, though available on Windows with support tools.
- **Protocol**: Operates over TCP/IP, typically on port **2049**

❖ How NFS Works:

1. A **server** exports one or more directories.
2. A **client** mounts those directories over the network.
3. The mounted directories appear as part of the client's file system.

✓ **The protocol handles**:

- File access (read/write)
- Directory traversal
- File locking (with appropriate version support)
- User authentication (in advanced versions)

Communication is typically done using Remote Procedure Calls (RPCs)

✓ Typical Use Cases

1. **Centralized File Storage :-** Multiple clients need access to a shared file system.
2. **Home Directory Sharing :-** Users' home directories are stored on a central server.
3. **Web Server Clustering :-** Multiple web servers share the same content (e.g., HTML, images, videos).
4. **Application Data Sharing :-** Applications (e.g., databases, analysis tools) share logs or datasets over NFS.
5. **Backup and Archiving :-** Backup servers mount file systems via NFS to copy data for backup.
6. **Virtual Machine Storage :-** Hypervisors or containers use NFS for persistent storage.
7. **Development Environments :-** Developers access and modify shared codebases or environments.

❖ What is Amazon EFS?

- **Amazon EFS** stands for **Amazon Elastic File System**. It's a **fully managed, scalable, and elastic network file system** offered by **AWS (Amazon Web Services)**, designed to be used with Amazon EC2 instances and other AWS services

✓ ***Key Features of Amazon EFS:***

- **Fully managed**
  - AWS handles provisioning, patching, scaling, and availability
- **Elastic**
  - Automatically scales up or down as files are added or removed
- **Shared access**
  - Multiple EC2 instances or containers can access the file system at the same time
- **Standard NFS support**
  - Compatible with NFSv4 and NFSv4.1 protocols
- **Durable and highly available**
  - Data is stored across multiple Availability Zones (AZs)
- **Secure :-** Support IAM, VPC security groups, encryption at rest/in transit

❖ Real-Time Use Cases of Amazon EFS

- **Use Case**
  - Real-Time Advantage
- **Web hosting**
  - Shared, live-updated content
- **Microservices (ECS/EKS)**
  - Shared config/logs at runtime
- **Media processing**
  - Concurrent read/write for large files
- **ML/Analytics pipelines**
  - Centralized, simultaneous data access
- **Developer environments**
  - Shared tools and codebases
- **Home directories**
  - Persistent user sessions
- **CI/CD pipelines**
  - Shared build/workspace folders

❖ Comparison Table: EFS vs EBS

Feature	**Amazon EFS**	**Amazon EBS**
Type	Network File System (shared file storage)	Block Storage (attached to a single instance)
Access	Multiple instances (shared via NFS)	One instance at a time (except multi-attach
Protocol	NFSv4.0 / NFSv4.1	Raw block-level I/O

Use Case	Shared storage for Linux-based workloads	Boot volumes, databases, low-latency apps
Scalability	Auto-scales up/down with usage	Fixed size; must resize manually
Durability & Availability	Multi-AZ, highly available	AZ-specific unless backed up
Performance Modes	General Purpose, Max I/O	Varies by volume type (gp3, io1, etc.)
Cost Model	Pay per GB used (plus throughput class	Pay per provisioned GB
Mounting Requirement	Mount via NFS client	Attach as a disk to EC2

❖ How does EFS handle scaling?

1. EFS **automatically scales** up or down based on the amount of data stored.
2. **No need** for manual provisioning or resizing.
3. It provides instant access to additional storage as needed

❖ What security measures can be implemented in Amazon EFS?

1. Use **IAM** (Identity and Access Management) **policies** to control access.
2. Attach **security groups** to allow traffic only from specific **EC2 instances**.
3. Enable encryption at rest using AWS Key Management Service (KMS).
4. Enable encryption in transit using **TLS** (Transport Layer Security).
5. Use AWS **VPC** and **NACLs** (Network ACLs) to restrict access.

❖ How can an EC2 instance mount an EFS file system?

1.Install the necessary NFS client:

sudo yum install -y amazon-efs-utils # For Amazon Linux 2 / RHEL

sudo apt install -y nfs-common # For Ubuntu/Debian

2.Create a mount point:

sudo mkdir -p /mnt/efs

3.Mount the EFS file system:

sudo mount -t efs fs-xxxxxxxx:/ /mnt/efs

4.To make it persistent, add an entry to /etc/fstab:

fs-xxxxxxxx:/ /mnt/efs efs defaults,_netdev 0 0

❖ What are the throughput modes in Amazon EFS?

- **Bursting Throughput Mode(Default):**

1. Throughput **automatically scales** with the size of your file system.
2. Through put rate You get **1 MiB/s per GiB** of data stored, up to a limit.

- **Provisioned Throughput Mode:-**

1. You manually set the throughput **independent** of the file system size.
2. Choose a **fixed throughput rate (in MiB/s)**, even if your data storage is small.

❖ How Is Amazon EFS Billed?

- Amazon EFS (Elastic File System) is billed based on **usage** and selected **performance (measured in GB/month).**
- **Optional costs** for provisioned throughput and backup storage

❖ What monitoring tools can be used for Amazon EFS?

- **Amazon CloudWatch:-** Provides metrics, logs, and alarms for EFS performance and usage.
- **AWS CloudTrail:-** Logs API calls related to EFS (e.g., create, delete, modify file systems).
- **Amazon Trusted Advisor:-** Provides optimization and security checks for EFS usage.